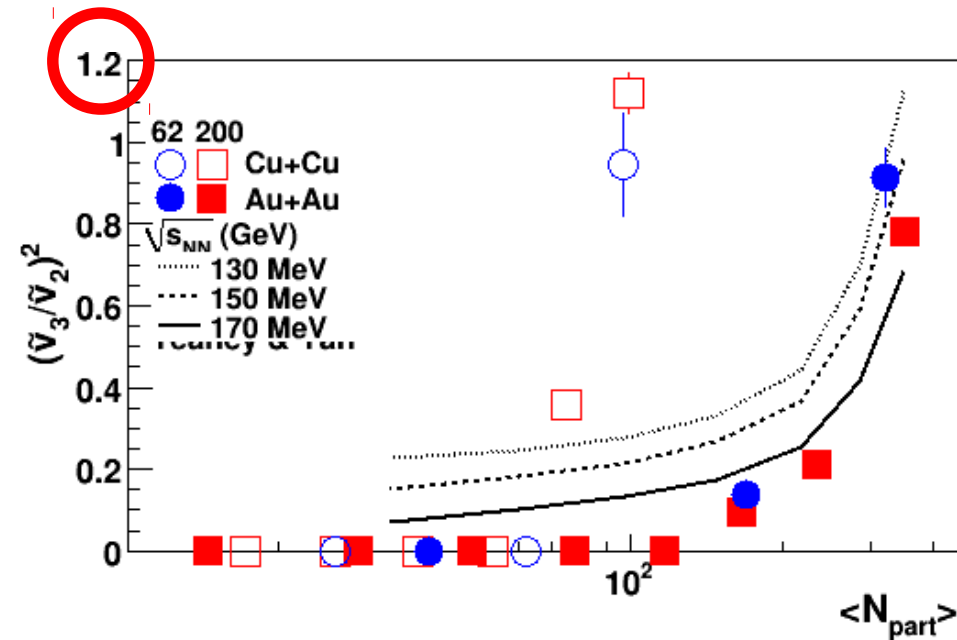


Collision system and energy dependence of di-hadron correlations

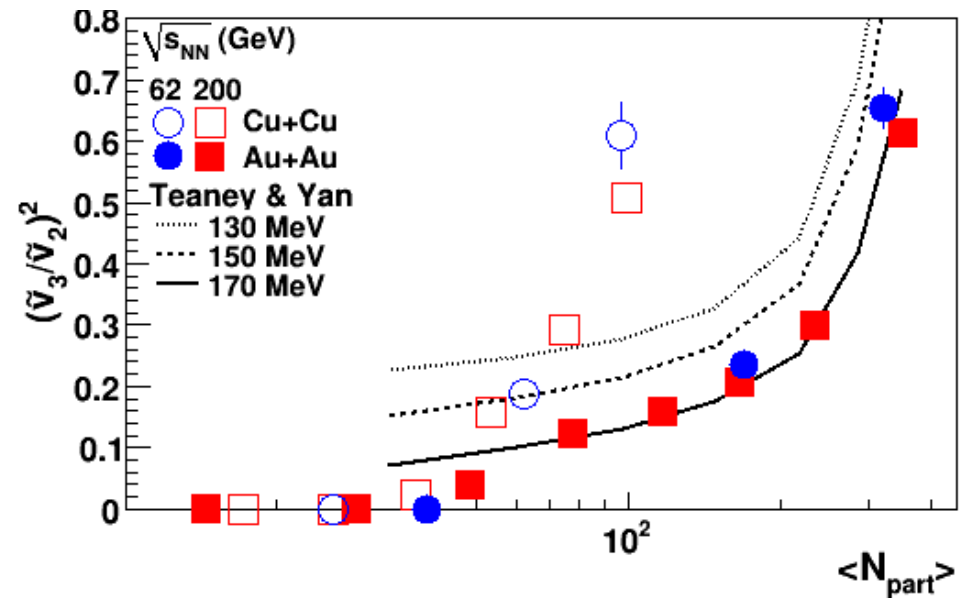
PAs: Christine Nattrass, Jana Bielcikova,
Helen Caines, Jörn Putschke

Note that in this presentation we use v_n interchangeably with $\tilde{v}_n$. This is not because we think that the Fourier components are necessarily flow but because this is for internal consumption only therefore we do not think that it is necessary to spend the time making the plots pretty.

Trying to fix v2



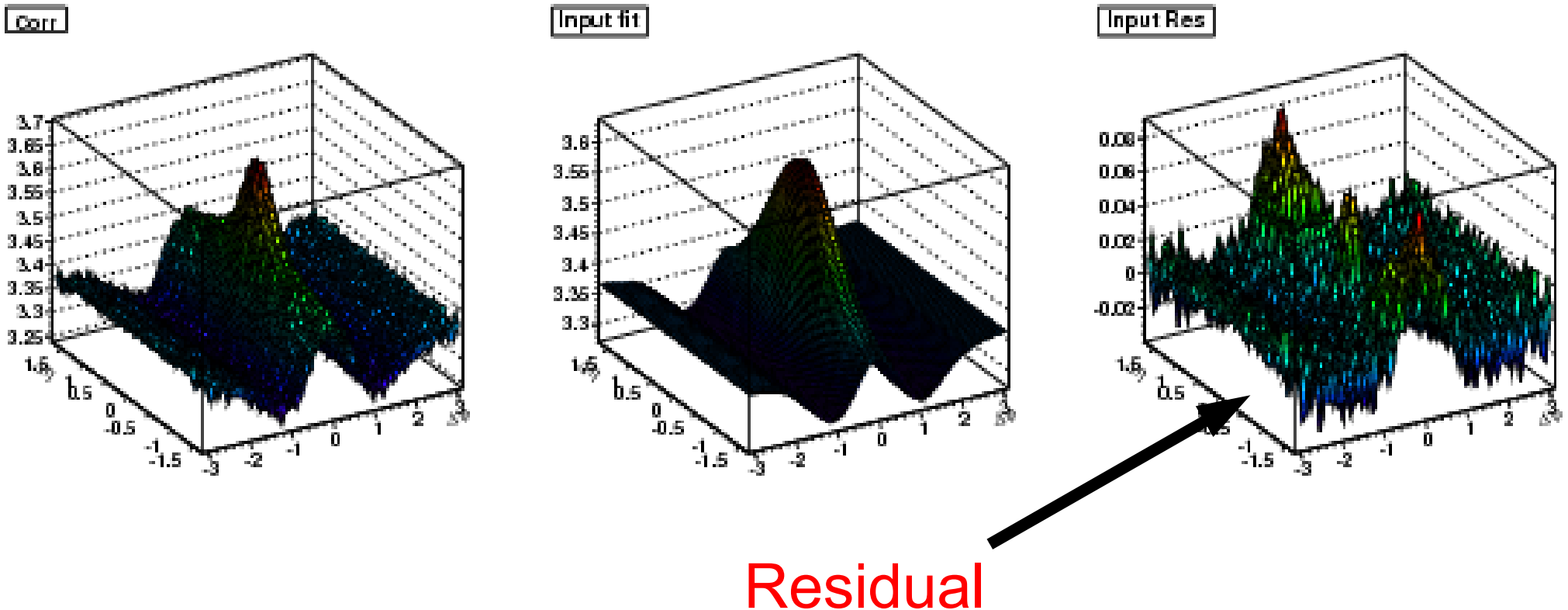
Fixed v2



Fit

- Same trend but different value
- However, *fit quality is really poor*

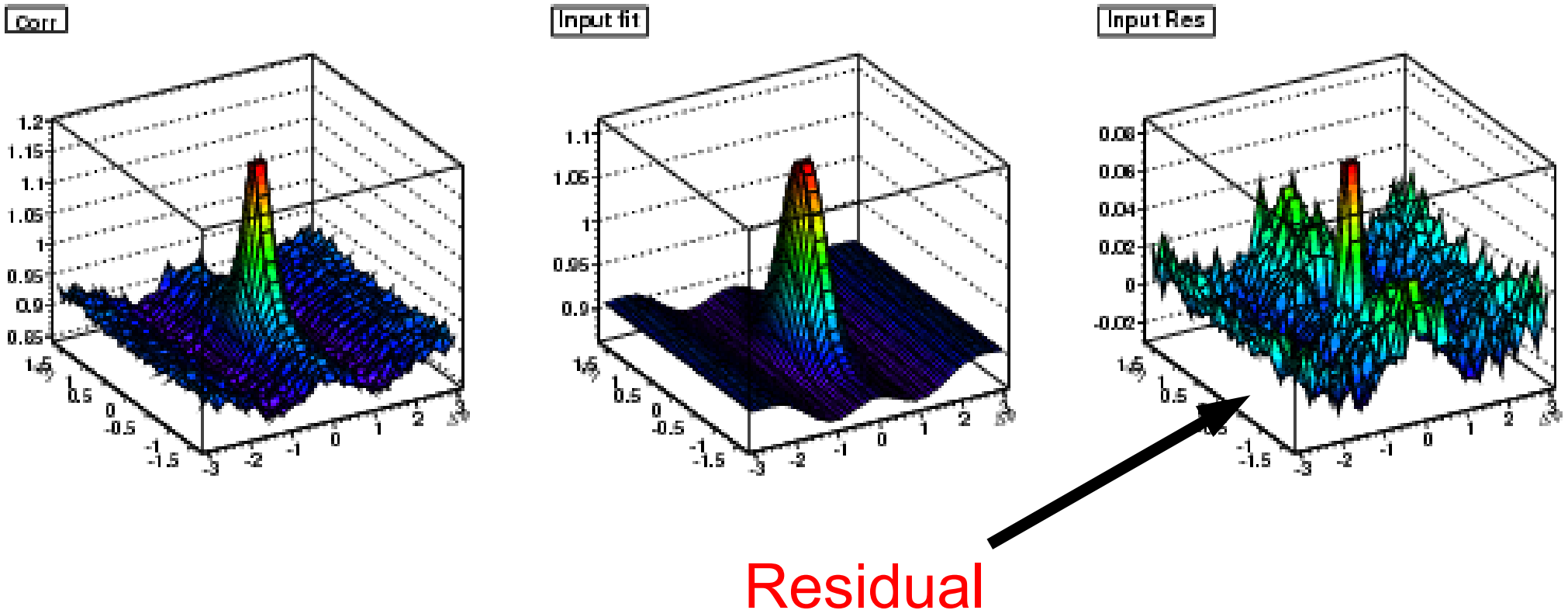
Fixing v2 – how poor?



Residual

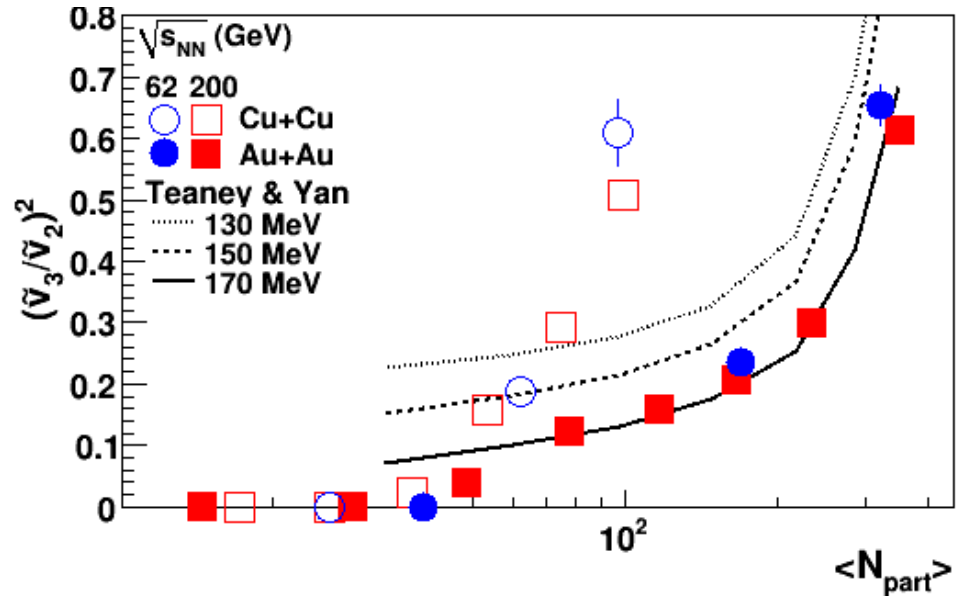
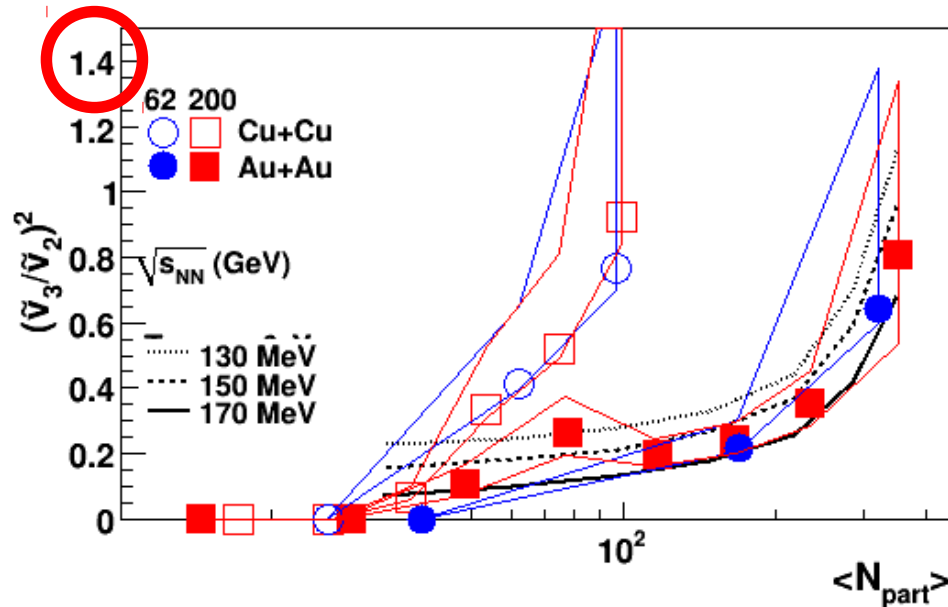
AuAu200 0-12%

Fixing v2 – how poor?



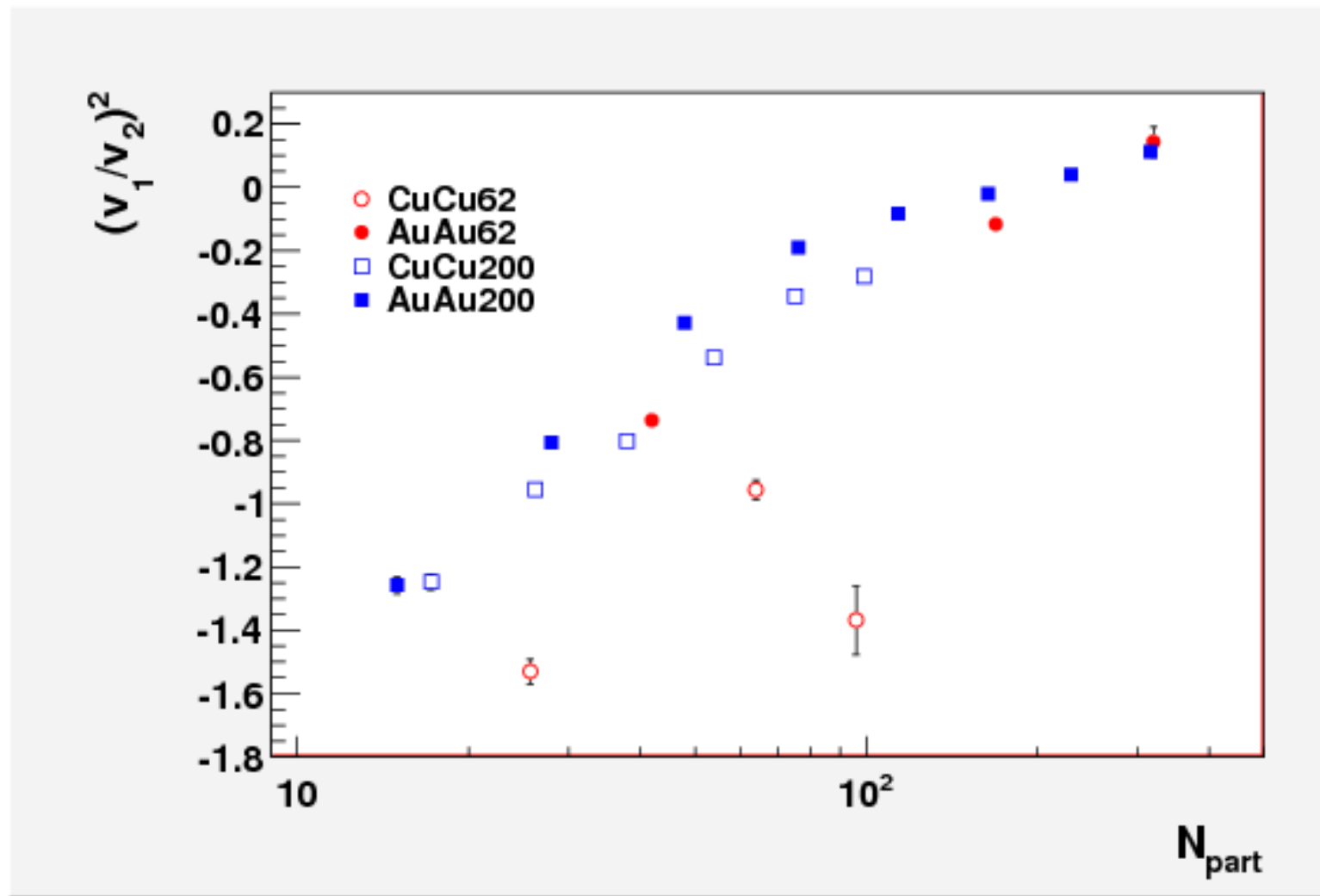
CuCu200 0-10%

Determine v_3 from fit and use standard v_2

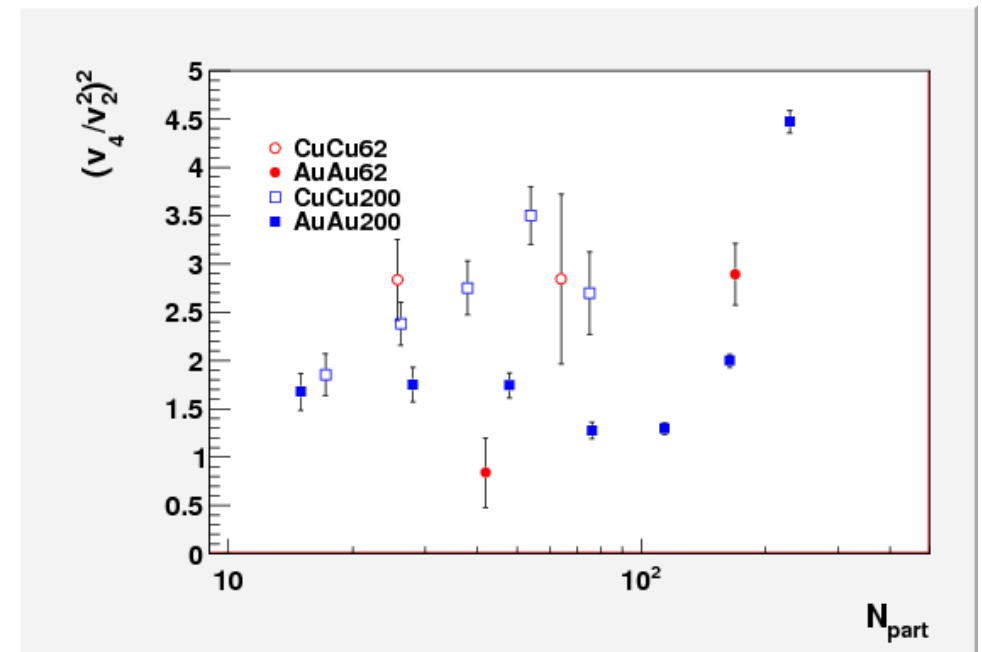
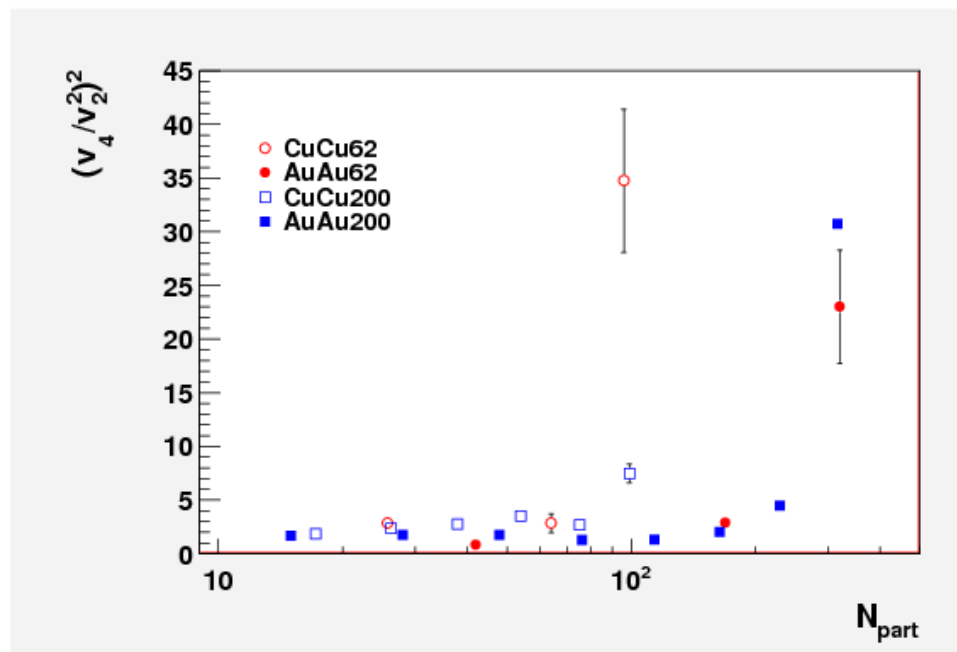


- Allows unambiguous calculation of some systematic error, at least due to v_2
- Two methods always agree when $v_3=0$, are within “error” for 0-40% Au+Au 200, all Au+Au 62, not within error for Cu+Cu (as expected)
- However, using the v_2 from other methods puts forward a stronger message that v_3 is unambiguously flow, which we were trying to avoid.
- **Proposal:** Keep the plot as is but add the v_3 from the fit to the table with v_2 information from the ZYAM method. This would allow people to interpret the data however they want and use whatever v_2 they want if they do not think we have made the correct choice.

$(v_1/v_2)^2$ from fit



$(v_4/v_2^2)^2$ from fit



$$v_2^2(\text{fit}) - v_2^2(\text{ZYAM})$$

